Ahmed Ben Haj Yahia

France +33 06 51 47 05 00
ahmed.belhajyahia@supcom.tn
Ahmed Ben haj yahia ahmedbhyy
Online Portfolio

Education

SUP'COM
Sept 2023 – Sept 2026

Higher School of Communication of Tunis *Telecommunications Engineering*

- Coursework: Signal Processing, AI, Cloud Computing, IoT, Networking

IPEIT
Sept 2021 – June 2023

Preparatory Institute for Engineering Studies of Tunis *Applied Sciences*

- National Rank: 147/1800
- Coursework: Mathematics, Mechanics, Physics, Chemistry, Algebra

Experience

IMT Nord Europe – CERI-SN
Apr 2026 – Sep 2026

Engineering Internship – Data Science & AI

- Analyzed energy data of a smart building (photovoltaic production, consumption, building management system) to identify inefficiencies
- Developed forecasting models for photovoltaic production and HVAC behavior using machine learning and deep learning
- Designed a multi-agent predictive control architecture to improve solar self-consumption and reduce energy purchased from the grid
- Evaluated the approach in simulation against the existing operation of the building

Freelancing
Jan 2024 – Present

Full-Stack Developer

- Developed a stock management and manufacturing order system for a protein production industry
- Built the *Errayen App*, a real estate management platform
- Created multiple e-commerce websites and event management solutions

Sagemcom Software & Technologies
2025

Engineering Internship – Artificial Intelligence

- Conducted an engineering internship focused on dynamic balancing of microgrids with energy storage using real-time data
- Designed and implemented AI-based models for energy management optimization
- Analyzed real-time data streams and evaluated system performance

Office de l'Aviation Civile et des Aéroports (OACA)
2024

Human Integration Internship

- Human integration internship aimed at understanding the professional environment and organizational structure of OACA
- Gained exposure to institutional workflows, teamwork, and professional practices

Projects

Customer Service Chatbot for Insurance (Tunisian Dialect)

- Developed a conversational AI chatbot designed for the insurance sector, capable of understanding and responding in the Tunisian dialect.
- Built a RAG-based system integrating natural language processing with a custom knowledge base to answer domain-specific queries.
- Implemented a multi-agent architecture using the Crew framework, enabling specialized agents to collaborate for improved response accuracy and task handling.
- Enhanced accessibility and customer support by bridging the language gap between clients and insurance services.

AI-Powered Financial News Automation System

- Developed a multi-agent system using the **CrewAI** framework (`Agent`, `Task`, `Crew`) to automate financial news analysis and reporting.
- Designed specialized agents for fetching the latest market news, summarizing articles, translating content into Arabic, and sending automated reports via email.
- Integrated **NewsAPI** for real-time financial data collection and **Brevo API** for automated email delivery.
- Utilized large language models through **OpenRouter** to power intelligent summarization and translation workflows.
- Demonstrated advanced orchestration between autonomous agents for streamlined financial information monitoring and communication.

AI Fitness Assistant with Memory Integration

- Developed an intelligent fitness assistant capable of personalized coaching and progress tracking.
- Integrated **SuperMemory API** for long-term memory storage and contextual recall of user history.
- Deployed the agent as a **Telegram bot** with real-time interaction, adaptive recommendations, and conversational goals.
- Combined **LangChain**, **OpenAI API**, and **function calling** for multi-step reasoning and dynamic goal execution.

Intelligent Energy Management System

- Developed system for real-time monitoring of production, storage, and consumption
- Optimized storage usage, reduced energy losses, and maintained grid stability

AI-Powered Virtual Dressing Room

- Mobile app using computer vision for virtual try-on with augmented reality
- Implemented deep learning models for image retrieval and pose estimation

General-Purpose ERP System

- Designed and developed a scalable, modular ERP system covering core business functions including sales, finance, inventory management, and human resources

Technical Skills

- **Languages:** Python, Dart, C++, JavaScript, SQL
- **Frontend:** Flutter, React
- **Backend:** Node.js, AWS (Lambda, S3, API Gateway, DynamoDB, EC2, SQS, SNS, Kinesis, SageMaker)
- **AI/ML:** TensorFlow, PyTorch, LightGBM, Scikit-learn
- **Computer Vision:** OpenCV, YOLOv8, Image Processing, Object Detection
- **LLMs & Agentic AI:** CrewAI, LangChain, OpenAI API, Hugging Face Transformers, RAG, Function Calling, Vector Databases (FAISS, Chroma)
- **DevOps:** Docker, Kubernetes, Ansible, Jenkins, Microservices, CI/CD

Awards & Achievements

IEEE WIE Empower Hackathon

- **1st Place** – Built an e-learning platform with teacher chatbot (web + mobile)

AI Anatomy Hackathon

- **1st Place** – Developed AI-driven healthcare platform with risk analysis tools

MAE MutualHack Hackathon

- **3rd Place** – Built hybrid learning platform with sign language recognition (YOLOv8, MediaPipe)

GothamHack Hackathon

- **2nd Place** – GuardianLock, Flutter app for car security via face recognition

Entrepreneurship Challenge, SUP'COM

- Designed and implemented an AI-driven mobile application enabling real-time interaction for deaf users through speech-to-text, text-to-speech, and intelligent language processing.

Club Activities

Junior Sup'Com Entreprise

Training Manager

- Organized mobile development trainings and mentored new developers

TEAM SUP'COM

Event Organizer

- Planned and organized student events, coordinated teams, managed logistics, and ensured smooth execution of activities.

Languages

- **Arabic:** Native speaker
- **English:** C1 level (TOEIC certified)
- **French:** C1 level (TCF certified)